

Assignment 7

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This report summarizes the results for well-log missing prediction using the dataset `data/WellLog/well14tutorial2.xlsx`. The input logs are DTCO, ECGR, RHOB, and PHIT, while the target log is DTSM. All models were trained in a supervised setting and evaluated by RMSE and R^2 , where lower RMSE and higher R^2 indicate better performance.

1. Optimal Hyperparameters for RNN, GRU, and LSTM

The hyperparameter search considered base hidden size $\{8, 16\}$, number of recurrent layers $\{2, 4\}$, learning rate $\{0.001, 0.0005\}$, batch size 128, and weight decay 0.0. For the original setting with sequence length 8, the best configuration in each model family was selected by the lowest validation RMSE.

Table 1: Best hyperparameters at sequence length 8

Model	Hidden Size	Layers	Learning Rate	Batch Size	Val. RMSE	Val. R^2
RNN	16	2	0.0010	128	17.7339	0.7715
GRU	16	4	0.0010	128	15.2362	0.8313
LSTM	16	2	0.0010	128	15.4917	0.8256

These results show a consistent preference for base hidden size 16 and learning rate 0.0010. GRU benefits from a deeper four-layer architecture, whereas RNN and LSTM achieve their best validation results with two recurrent layers.

2. Test Results Evaluated by RMSE and R^2

After selecting the best hyperparameters, the unidirectional models were evaluated on the test set under the original sequence length 8.

Table 2: Test performance of unidirectional models at sequence length 8

Model	RMSE	R^2
RNN	12.1102	0.8802
GRU	11.2701	0.8962
LSTM	10.6406	0.9075

Among the three baseline models, LSTM performs best because it produces the lowest RMSE and the highest R^2 . GRU ranks second, while RNN gives the weakest test performance in the original sequence-length setting.

3. Replace All Layers with Bidirectional Layers and Re-evaluate

All recurrent layers were replaced by their bidirectional counterparts and then re-evaluated using sequence length 8.

Table 3: Test performance of bidirectional models at sequence length 8

Model	RMSE	R^2
BiRNN	11.9695	0.8830
BiGRU	12.3207	0.8760
BiLSTM	10.1331	0.9161

Compared with the unidirectional baselines, bidirectionality gives mixed effects:

- RNN improves slightly from RMSE 12.1102 to 11.9695, and R^2 rises from 0.8802 to 0.8830.
- GRU becomes worse after the bidirectional substitution, with RMSE increasing from 11.2701 to 12.3207 and R^2 dropping from 0.8962 to 0.8760.
- LSTM benefits the most: RMSE improves from 10.6406 to 10.1331, and R^2 increases from 0.9075 to 0.9161.

Therefore, bidirectional layers are helpful for RNN and especially LSTM on this dataset, but they do not help GRU in the original sequence-length setting.

4. Compare All Models and Determine the Best-performing Model

Table 4 compares all tested models under the fixed setting of sequence length 8, including both unidirectional and bidirectional architectures.

Table 4: Ranking of all evaluated models at sequence length 8

Model	RMSE	R^2
BiLSTM	10.1331	0.9161
LSTM	10.6406	0.9075
GRU	11.2701	0.8962
BiRNN	11.9695	0.8830
RNN	12.1102	0.8802
BiGRU	12.3207	0.8760

When only sequence length 8 is considered, the best-performing model is **BiLSTM**. Its hyperparameters are base hidden size 16, number of layers 2, learning rate 0.0010, and batch size 128. This model is selected for three reasons:

- It achieves the lowest RMSE among all models at sequence length 8, namely 10.1331.
- It also achieves the highest R^2 , namely 0.9161, so it performs best on both evaluation metrics.
- The second-best model is the unidirectional LSTM, but its RMSE is still 0.5075 higher than BiLSTM.

This result indicates that, within the sequence length 8 setting, the bidirectional LSTM captures temporal information more effectively than the other recurrent architectures. In particular, it improves over the unidirectional LSTM baseline and clearly outperforms both RNN-based and GRU-based alternatives.

5. Change the Sequence Length to 16 and Re-evaluate the Models

After increasing the sequence length from 8 to 16, the best hyperparameters for the three unidirectional model families remain similar, and the validation-optimal settings are shown below.

Table 5: Best hyperparameters at sequence length 16

Model	Hidden Size	Layers	Learning Rate	Batch Size	Val. RMSE	Val. R^2
RNN	16	2	0.0010	128	12.6448	0.8769
GRU	16	4	0.0010	128	11.1881	0.9036
LSTM	16	2	0.0010	128	11.2006	0.9034

Table 6: Test performance after changing sequence length to 16

Model	RMSE	R^2
RNN	9.3922	0.9301
GRU	6.6457	0.9650
LSTM	6.7129	0.9643
BiRNN	8.0919	0.9481
BiGRU	7.1265	0.9598
BiLSTM	6.3133	0.9684

The change from sequence length 8 to 16 improves every tested model:

- Unidirectional RNN improves from RMSE 12.1102 to 9.3922, while R^2 increases from 0.8802 to 0.9301.
- Unidirectional GRU improves from RMSE 11.2701 to 6.6457, while R^2 increases from 0.8962 to 0.9650.
- Unidirectional LSTM improves from RMSE 10.6406 to 6.7129, while R^2 increases from 0.9075 to 0.9643.
- Bidirectional RNN improves from RMSE 11.9695 to 8.0919, while R^2 increases from 0.8830 to 0.9481.
- Bidirectional GRU improves from RMSE 12.3207 to 7.1265, while R^2 increases from 0.8760 to 0.9598.
- Bidirectional LSTM improves from RMSE 10.1331 to 6.3133, while R^2 increases from 0.9161 to 0.9684.

This consistent improvement suggests that a longer temporal context helps the networks capture the relationships needed for DTSM prediction. After the re-evaluation, the strongest configuration remains BiLSTM with sequence length 16.

Conclusion

For this well-log missing prediction task, the most reliable hyperparameter pattern is hidden size 16, learning rate 0.0010, and batch size 128, with GRU preferring four recurrent layers and RNN/LSTM preferring two layers. Bidirectional modeling helps RNN and LSTM, but not GRU, and increasing the sequence length to 16 improves every tested architecture. The final recommended model is therefore **BiLSTM with sequence length 16**.

Code

`https://github.com/WKKKKKKKKKKKKKK/ML\_IN\_GEO`